

Problem H – Hidden Symmetry of Valdris

Deep beneath the Citadel of Eternal Letters lies the Whispering Archives — a vault sealed for millennia, whose enchanted doors open only to those who can recite two simultaneous palindromic incantations.

The vault’s guardian, Valdris the Scribe, inscribed its secrets across two sacred scrolls: A and B . To unseal the vault, a supplicant must pick a split point i and a meeting index j , then chant both of the following at once:

$$P_1 = A[1 \dots i] + B[j \dots |B|]$$
$$P_2 = B[1 \dots i] + A[j \dots |A|]$$

Both P_1 and P_2 must be palindromes simultaneously. Centuries of failed attempts have filled the antechamber with very flat scholars. Help the next adventurer survive by counting every valid pair (i, j) such that $1 \leq i, j \leq |A|$.

Input

The first line contains T ($1 \leq T \leq 10^5$).

Each test case consists of two lines containing strings A and B of lowercase English letters, with $|A| = |B|$ ($1 \leq |A|, |B| \leq 10^6$).

Sum of $|A| + |B|$ over all test cases is less than 2×10^6

Output

For each test case, a single integer — the number of valid pairs (i, j) .

Sample input 1	Sample output 1
4	25
abacaba	9
abacaba	7
aaa	10
aaa	
abc	
cba	
abyyycd	
dczzzba	

Notes

A string S of length ℓ is a palindrome if $S[k] = S[\ell - k + 1]$ for all $1 \leq k \leq \ell$. The empty string is considered a palindrome.